RESEARCH MONOGRAPH

ON

CYBERCRIME AND DIGITAL EVIDENCE IN BANGLADESH: PROBLEM AND PROSPECT

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ID: 2243291009

THE THESIS SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE DEGREE OF LL.M.

DEPARTMENT OF LAW

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DECLARATION

I hereby declare that the thesis entitled **"CYBERCRIME AND DIGITAL EVIDENCE IN BANGLADESH: PROBLEM AND PROSPECT.''** is my own original work and has been written by me in its entirety. I have not plagiarized any part of this work and have duly acknowledged all sources of information, data, ideas, or quotations by citing and referencing them appropriately according to academic standards.

This thesis has not been submitted, in whole or in part, for any other degree or academic qualification at this or any other institution. Any contributions made by others to this research, including guidance, resources, or collaborative support, have been fully credited and acknowledged.

I take full responsibility for the content and integrity of this thesis.

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This thesis stands as a reflection of personal effort and the collective guidance, inspiration, and support I have received from numerous individuals and institutions.

List of Abbreviations

1. **AI** – Artificial Intelligence
2. **API** – Application Programming Interface
3. **BCS** – Bangladesh Computer Council
4. **BTRC** – Bangladesh Telecommunication Regulatory Commission
5. **CA** – Certifying Authority
6. **CC** – Cybercrime
7. **CCTV** – Closed-Circuit Television
8. **CERT** – Computer Emergency Response Team
9. **CII** – Critical Information Infrastructure
10. **CIRT** – Computer Incident Response Team
11. **CID** – Criminal Investigation Department
12. **CTTC** – Counter Terrorism and Transnational Crime Unit
13. **CSIRT** – Computer Security Incident Response Team
14. **CSW** – Cyber Support for Women
15. **CSA** – Cyber Security Act
16. **CTA** – Counter Terrorism Act
17. **DDoS** – Distributed Denial of Service
18. **DISA** – Digital Security Agency
19. **DSA** – Digital Security Act
20. **DNS** – Domain Name System
21. **ECA** – Evidence Collection and Analysis
22. **E-Commerce** – Electronic Commerce
23. **FBI** – Federal Bureau of Investigation
24. **GCI** – Global Cybersecurity Index
25. **GDPR** – General Data Protection Regulation
26. **GoB** – Government of Bangladesh
27. **ICT** – Information and Communication Technology
28. **ICT Act** – Information and Communication Technology Act, 2006
29. **ID** – Identification

30. **IP** – Internet Protocol
31. **ISP** – Internet Service Provider
32. **IT** – Information Technology
33. **ITU** – International Telecommunication Union
34. **LEA** – Law Enforcement Agency
35. **MLA** – Mutual Legal Assistance
36. **MoICT** – Ministry of Information and Communication Technology
37. **NCC** – National Cybersecurity Center
38. **NCCC** – National Cybercrime Coordination Center
39. **NIST** – National Institute of Standards and Technology
40. **OECD** – Organisation for Economic Co-operation and Development
41. **PBI** – Police Bureau of Investigation
42. **PCSW** – Police Cyber Support for Women
43. **SOC** – Security Operations Center
44. **SMS** – Short Message Service
45. **SSL** – Secure Sockets Layer
46. **ToR** – Terms of Reference
47. **UN** – United Nations
48. **UNESCO** – United Nations Educational, Scientific and Cultural Organization
49. **UNODC** – United Nations Office on Drugs and Crime
50. **VPN** – Virtual Private Network

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Abstract

Cybercrime has emerged as a significant threat to national security, economic stability, and individual privacy in Bangladesh, coinciding with rapid digitalization and widespread internet adoption. This research examines the current landscape of cybercrime in Bangladesh and the challenges associated with managing digital evidence for effective investigation and prosecution. The study explores the legal framework, including the Digital Security Act 2018, ICT Act 2006, and amended Evidence Act 2022, alongside institutional responses from agencies such as CID, CTTC, and the Digital Security Agency. Key issues identified include legal ambiguities, procedural gaps in digital evidence handling, limited forensic capacity, inadequate inter-agency coordination, and low public awareness. The paper also analyzes prospects for reform, highlighting the need for comprehensive digital evidence legislation, capacity building, international cooperation, and public education. Strengthening these areas is essential for a secure digital environment and an effective cybercrime justice system in Bangladesh.

Keywords

Cybercrime, Digital Evidence, Bangladesh, Digital Security Act, ICT Act, Cyber Forensics, Law Enforcement, Legal Framework, Cyber Law, Data Protection

CHAPTER 1

INTRODUCTION

1.1 Introduction

In the era of globalization and rapid technological advancement, Bangladesh is undergoing a transformative shift toward digitalization. Internet access, mobile banking, social media usage, e-commerce platforms, and online government services have expanded rapidly under national initiatives such as "Digital Bangladesh" and the emerging vision of "Smart Bangladesh 2041." While these developments have opened new pathways for economic growth, governance improvement, and social connectivity, they have also given rise to significant risks and vulnerabilities in cyberspace.

Cybercrime has emerged as a major threat affecting individuals, financial institutions, corporations, and the state itself. Offenses such as online fraud, mobile financial scams, identity theft, hacking, ransomware, phishing, child exploitation, cyberbullying, defamation, and attacks on critical information infrastructures are increasingly common. These crimes often occur anonymously, cross borders within seconds, and leave little trace without proper technical tools.

In this context, **digital evidence**, which includes all electronically stored or transmitted information, has become critical for detecting, investigating, and prosecuting cybercrimes. Bangladesh has made notable progress—such as developing forensic labs under the CID and CTTC, amending the Evidence Act 1872 to include electronic evidence, and enacting laws like the Cyber Security Ordinance, 2025[1]. However, significant gaps remain in legal frameworks, forensic capacity, admissibility standards, interagency coordination, digital literacy, and cross-border data access.

This chapter introduces the topic, outlines the research problem, explains the rationale and significance of the study, and presents the research questions, objectives, scope, methodology, and structure of the research paper.

[1] This ordinance, which was issued as a gazette in May 2025, replaces the previous Cyber Security Act of 2023 (CSA). The 2023 CSA had, in turn, replaced the controversial Digital Security Act (DSA) of 2018.

1.2 Background of the Study

Bangladesh's digital ecosystem has expanded rapidly over the past decade. Government services, banking systems, educational institutions, and private sectors increasingly depend on digital platforms. As digital dependency increases, so do opportunities for cybercriminals. The Bangladesh Bank cyber heist (2016)[2], massive financial scams via mobile banking, and frequent incidents of online harassment illustrate the scale of the problem.

Digital evidence now plays a decisive role in modern criminal investigations. Everything from emails, call logs, metadata, IP addresses, CCTV footage, cloud storage, browser history, and mobile data contributes to establishing factual truth. Yet, digital evidence is fragile, easily alterable, and often stored in foreign jurisdictions, making its collection and authentication difficult.

Although laws exist, many investigators and prosecutors lack specialized training. Courts sometimes hesitate to accept digital evidence when the chain of custody is unclear. The gap between technological change and legal readiness necessitates a systematic academic inquiry.

1.3 Problem Statement

Despite rapid digitalization, Bangladesh faces a series of structural and operational challenges in combating cybercrime and handling digital evidence. The primary problems include:

- Insufficient legal clarity and overlapping laws
- Limited forensic capacity and lack of technical expertise
- Inadequate chain-of-custody management
- Cross-border data collection barriers
- Insufficient coordination among law enforcement, regulators, and judiciary
- Low public awareness and weak cybersecurity culture

These challenges undermine effective investigation, prosecution, and prevention of cybercrime.

[2] https://en.wikipedia.org/wiki/Bangladesh_Bank_robbery

1.4 Research Questions

The study is guided by three core research questions:

1. **What are the major forms, patterns, and legal responses to cybercrime in Bangladesh?**
2. **What challenges do law enforcement agencies and courts face in collecting, preserving, and presenting digital evidence?**
3. **What prospects and opportunities exist for strengthening cybercrime governance and digital evidence management in Bangladesh?**

1.5 Objectives of the Study

General Objective

To analyze the problems and prospects of cybercrime management and digital evidence in Bangladesh.

Specific Objectives

- To examine the nature and trends of cybercrime in Bangladesh.
- To evaluate the effectiveness of current legal and institutional frameworks.
- To identify challenges in digital evidence preservation and forensic investigation.
- To explore prospects for improvement through legal reforms, capacity-building, and technological innovation.

1.6 Scope of the Study

The study focuses on:

- Major cybercrimes affecting individuals, businesses, and national security

- Relevant laws such as the ICT Act 2006[3], Cyber Security Ordinance, 2025, Evidence Act 1872 (amended 2025), and the Telecommunication Act
- Operational capacities of agencies like CID, CTTC, BTRC, Bangladesh Bank, and judiciary
- Forensic challenges including data integrity, admissibility, and chain of custody
- Prospects for modernization in line with global standards

The research is limited to Bangladesh but considers international practices for comparative analysis.

1.7 Significance of the Study

Cybercrime threatens public safety, financial stability, privacy, social harmony, and national security. Understanding the complexities of cyber-offenses and digital evidence is essential for:

- Policymakers designing effective laws
- Law enforcement agencies improving investigation capacity
- Courts evaluating electronic evidence more effectively
- Researchers and students studying cyberlaw and forensics

The study contributes to building a secure digital environment essential for Smart Bangladesh.

1.8 Methodology

The study uses a **qualitative, doctrinal, and socio-legal research method**, involving:

- Analysis of statutes, rules, judicial decisions, and international instruments
- Review of academic writings, research reports, and forensic guidelines
- Examination of official data from law enforcement agencies
- Conceptual and comparative analysis

This approach allows a comprehensive understanding of both legal and technical dimensions.

[3] http://bdlaws.minlaw.gov.bd/act-details-950.html

1.9 Limitations of the Study

- Official cybercrime statistics are limited and sometimes inconsistent
- Some forensic technologies are confidential
- Rapid technological evolution means some findings may become time-sensitive
- Limited accessibility to international cybercrime investigation data

  Despite these constraints, the study relies on the best available information.

1.10 Conclusion

Cybercrime poses a multidimensional challenge to Bangladesh's ongoing digital transformation. While digital evidence offers crucial opportunities for modern criminal justice, managing it requires strong legal, institutional, and technical frameworks. This chapter has provided the foundation for the study by outlining the background, significance, research questions, objectives, methodology, scope, and limitations.

The next chapter will present the **conceptual and legal framework** of cybercrime and digital evidence, highlighting the theoretical foundations and existing legal instruments in Bangladesh.

CHAPTER 2

CONCEPTUAL AND LEGAL FRAMEWORK OF CYBERCRIME AND DIGITAL EVIDENCE

2.1 Introduction

Cybercrime and digital evidence are interconnected concepts that form the backbone of modern criminal justice systems. As Bangladesh experiences rapid technological expansion, understanding these concepts and the legal frameworks surrounding them becomes crucial. This chapter explores the definitions, classifications, and theoretical perspectives of cybercrime and digital evidence. It further analyzes national and international legal instruments governing these issues. By establishing a conceptual and legal foundation, this chapter supports the later analysis of challenges and prospects within the Bangladeshi context.

2.2 Concept of Cybercrime

2.2.1 Definition

Cybercrime refers to criminal activities conducted using computers, digital devices, or networks. The Council of Europe's **Budapest Convention (2001)**—the first international treaty on cybercrime—defines it as offenses against the confidentiality, integrity, and availability of computer systems, data, and related misuse.

In Bangladesh, neither the ICT Act 2006 nor the Digital Security Act 2018 provides a single comprehensive definition. Instead, both laws specify categories of offenses such as hacking, fraud, identity theft, cyberterrorism, and defamation. Therefore, cybercrime in Bangladesh is understood broadly as **any unlawful act committed using digital technology or targeting digital systems**.

2.3 Types and Classification of Cybercrime

Cybercrimes in Bangladesh may be classified as follows:

2.3.1 Crimes Against Individuals

- Cyberbullying, harassment, stalking
- Identity theft and impersonation
- Personal data theft
- Dissemination of intimate images
- Online defamation
- Financial scams through mobile banking

2.3.2 Crimes Against Property

- Hacking and unauthorized access
- Digital fraud and phishing
- Ransomware attacks
- Intellectual property theft
- Website defacement

2.3.3 Crimes Against the State

- Cyberterrorism
- Attacks on critical information infrastructure
- Digital propaganda and extremist recruitment
- Threats to national security databases

2.3.4 Hybrid Cybercrimes

Some crimes—such as human trafficking, money laundering, drug sales, and arms trafficking—use digital platforms as tools, creating hybrid forms of cyber-enabled crime.

2.4 Concept of Digital Evidence

2.4.1 Definition

Digital evidence refers to **any information stored, transmitted, or processed in digital form** that can be used in legal proceedings. This includes:

- Emails, SMS, call logs
- IP addresses and server logs
- CCTV footage and photographs
- Metadata
- Social media posts and chats
- Hard drives, USBs, mobile phones
- Cloud-stored data

Digital evidence is often volatile, easily altered, and technically complex, requiring specialized tools for extraction, preservation, and authentication.

2.5 Characteristics of Digital Evidence

- **Fragile:** Can be changed or deleted easily
- **Replicable:** Can be copied without altering the original
- **Volume-heavy:** Massive quantities of information
- **Cross-border:** Often stored in foreign jurisdictions
- **Metadata-driven:** Includes background information such as timestamps and geolocation

These characteristics require strict forensic protocols to ensure admissibility in court.

2.6 Chain of Custody in Digital Evidence

The chain of custody refers to the documented chronological record of the handling, storage, and transfer of digital evidence. It ensures:

- Authenticity
- Integrity
- Reliability
- Admissibility

In Bangladesh, agencies like **CID Digital Forensics Lab**, **CTTC Cyber Crime Unit**, and **Police Bureau of Investigation (PBI)** maintain chain-of-custody forms, although practices vary.

2.7 Legal Framework Governing Cybercrime and Digital Evidence in Bangladesh

2.7.1 Information and Communication Technology Act 2006 (ICT Act)

- First law addressing cyber offenses
- Penalized hacking, fraud, and unauthorized access
- Criticized for broad language and harsh penalties
- Many provisions replaced by the Digital Security Act 2018

2.7.2 Cyber Security Ordinance 2025[4]

- Primary law for cybercrime
- Covers offenses like hacking, digital fraud, pornography, defamation, and propaganda
- Provides authority for search, seizure, and forensic examination
- Criticized for vague provisions and potential misuse
- Currently undergoing review for reform

2.7.3 Evidence Act 1872 (Amendment 2025)

- Recognizes electronic records as admissible evidence

[4] [4] This ordinance, which was issued as a gazette in May 2025, replaces the previous Cyber Security Act of 2023 (CSA). The 2023 CSA had, in turn, replaced the controversial Digital Security Act (DSA) of 2018.

- Includes emails, digital documents, audio-video recordings, and digital signatures
- Requires proof of authenticity and integrity

2.7.4 Bangladesh Telecommunication Act 2001

- Governs telecom operators' data handling
- Grants BTRC authority over data interception and monitoring

2.7.5 Penal Code 1860

Although not designed for cybercrime, some sections indirectly apply—for example, cheating, fraud, extortion, and defamation conducted online.

2.7.6 Money Laundering Prevention Act & Anti-Terrorism Act

Used when cybercrime is linked with financial or extremist activities.

2.8 Institutional Framework in Bangladesh

2.8.1 Law Enforcement Agencies

- **CID (Cyber Forensics Lab)**
- **CTTC (Cyber Crime Unit, Dhaka Metropolitan Police)**
- **PBI (Cyber Investigation Unit)**
- **RAB Cyber Crime Unit**

These agencies investigate cybercrime, conduct forensic analysis, and support prosecution.

2.8.2 Regulatory Bodies

- **BTRC:** telecom regulation, data requests
- **Bangladesh Bank:** oversight of financial cybercrimes
- **NCC (National Cybersecurity Centre)** under Digital Security Agency

2.8.3 Judiciary

- Admissibility of digital evidence decided by trial courts
- Judges increasingly rely on expert forensic testimony

2.9 International Legal Instruments

Although Bangladesh is not yet a signatory to the **Budapest Convention**, its principles guide domestic reforms. Other influential instruments include:

- **UN General Assembly Resolutions on Cybersecurity**
- **ITU Guidelines on Cybercrime Legislation**
- **Interpol Global Cybercrime Strategy**
- **ASEAN & SAARC cyber cooperation models**

International cooperation is crucial for dealing with cross-border data requests and global cybercriminal networks.

2.10 Challenges in the Legal Framework

Key challenges include:

- Lack of unified and technologically updated laws
- Overlapping mandates among agencies
- Limited procedural clarity for digital evidence
- Absence of international data-sharing agreements
- Concerns over privacy and misuse of search powers

These issues will be explored in depth in Chapter 3.

2.11 Conclusion

This chapter has outlined the conceptual foundations and legal framework of cybercrime and digital evidence in Bangladesh. It highlighted the definitions, types, and characteristics of cybercrime, explained the nature and challenges of digital evidence, and examined the relevant legal and institutional structures. While Bangladesh has made progress in developing cyber laws and forensic capacities, several gaps remain in ensuring consistency, clarity, and international alignment.

CHAPTER 3

PROBLEMS AND CHALLENGES IN CYBERCRIME INVESTIGATION AND DIGITAL EVIDENCE MANAGEMENT

3.1 Introduction

Despite significant progress in digital transformation, Bangladesh continues to face major difficulties in effectively investigating cybercrime and managing digital evidence. Cybercrime is inherently complex—often transnational, technologically sophisticated, and rapidly evolving. In contrast, the institutional, legal, and technical capacities of law enforcement, regulators, and the judiciary have not kept pace with this evolution. As a result, detection rates remain low, digital evidence is frequently compromised, and many cases fail in court due to procedural or technological weaknesses.

This chapter analyzes the key problems and challenges confronting Bangladesh in dealing with cybercrime and digital evidence. These limitations are categorized as legal, technical, institutional, and socio-cultural, providing a comprehensive understanding of the obstacles that hinder effective cybercrime governance.

3.2 Legal Challenges

3.2.1 Lack of a Unified and Updated Legal Framework

Bangladesh's cyber laws are fragmented across multiple instruments—the Digital Security Act 2018, Telecommunication Act 2001, Penal Code 1860, Evidence Act 1872 (as amended), and sector-specific guidelines. This creates overlapping jurisdictions, inconsistent definitions, and confusion among investigators and prosecutors.

3.2.2 Ambiguities in Cybercrime Definitions

Many offenses, such as "defamation," "offensive content," and "propaganda," lack precise definitions. Vague language leads to misuse, wrongful arrests, and challenges in court due to inadequate legal clarity.

3.2.3 Admissibility Challenges of Digital Evidence

Although the Evidence Act now recognizes digital evidence, the requirements for admissibility—authentication, integrity, metadata verification—are often not followed due to lack of clear procedural rules.

3.2.4 Lack of International Cybercrime Agreements

Bangladesh is not a signatory to the Budapest Convention and lacks robust Mutual Legal Assistance Treaties (MLATs). As a result, obtaining digital data from foreign companies (Meta, Google, WhatsApp, etc.) is slow, difficult, and sometimes impossible.

3.3 Technical Challenges

3.3.1 Limited Digital Forensic Capacity

Forensic labs in CID, CTTC, and PBI have improved, but capacity remains insufficient. Backlogs are common due to:

- Shortage of forensic experts
- Outdated or limited forensic tools
- Slow processing of mobile and computer devices
- Lack of automation and advanced analytics

3.3.2 Vulnerability of Digital Evidence

Digital evidence is extremely fragile. Improper handling during seizure, storage, or transfer often leads to:

- Alteration
- Loss
- Corruption
- Metadata changes

This damages the chain of custody and weakens cases in court.

3.3.3 Encryption and Anonymity

End-to-end encrypted platforms (WhatsApp, Signal, Telegram) make it difficult to trace communications. Cybercriminals use VPNs, proxies, Tor browsers, and dark web services, making attribution extremely challenging.

3.3.4 Rapid Technological Change

Cybercrime technologies evolve faster than the capacity of law enforcement to adapt. Tools used by criminals—ransomware kits, phishing templates, botnets—are easily accessible and frequently updated.

3.4 Institutional Challenges

3.4.1 Fragmented Agency Coordination

CID, CTTC, PBI, RAB, BTRC, and Bangladesh Bank all deal with cybercrime, but coordination is often weak. Issues include:

- Jurisdictional overlap
- Delayed information sharing
- Lack of unified command and control
- Multiple investigations of the same case

3.4.2 Shortage of Skilled Human Resources

Most police officers and prosecutors lack specialized training in:

- Network forensics
- Malware analysis
- OSINT (Open-Source Intelligence)
- Chain of custody management
- Courtroom presentation of digital evidence

This contributes to delays and weak prosecutions.

3.4.3 Inadequate Judicial Understanding

Many judges remain unfamiliar with:

- Technical processes of evidence extraction
- Metadata interpretation
- Hash values and integrity verification
- Cyber forensic terminology

This sometimes leads to inconsistent judicial decisions.

3.4.4 Limited Resources and Budget

Cybercrime units require continuous investment in:

- Forensic software
- Hardware upgrades
- Secure storage facilities
- Training programs

Budget constraints hinder modernization.

3.5 Procedural Challenges

3.5.1 Weak Chain of Custody Practices

Failure to document each step—seizure, transfer, analysis—often causes evidence to be rejected in court. Many field officers do not use standardized chain-of-custody forms.

3.5.2 Delayed Reporting and Investigation

Victims frequently report cybercrimes late due to stigma, fear, or lack of awareness. Delayed reporting reduces the chances of retrieving original evidence.

3.5.3 Lack of Forensic Standardization

Bangladesh lacks national forensic standards equivalent to NIST (U.S.) or ENISA (EU). As a result, forensic practices vary across agencies, causing inconsistencies.

3.5.4 Handling of Mobile Financial Crimes

Mobile banking platforms (bKash, Nagad, Rocket) face widespread fraud. Investigating these crimes requires real-time data access, but bureaucratic delays hinder quick response.

3.6 Socio-Cultural and Behavioral Challenges

3.6.1 Low Public Awareness

Many citizens are unaware of:

- Cybersecurity practices
- Reporting procedures
- Legal protections
- Risks of oversharing personal information

This makes them easy targets for fraud, harassment, and identity theft.

3.6.2 Social Stigma

Victims of cyber harassment—especially women—avoid reporting due to fear of social judgment. This encourages offenders to continue criminal activities.

3.6.3 Rapid Digital Adoption Without Safety Culture

People often adopt technology without understanding security measures, resulting in increased vulnerability.

3.7 Cross-Border and Global Challenges

3.7.1 Jurisdictional Limitations

Cybercrime often involves servers, perpetrators, or victims located in different countries. National police cannot investigate beyond borders without international cooperation.

3.7.2 Slow MLAT Process

Obtaining data from foreign digital service providers may take months or years.

3.7.3 Absence of a Global Enforcement Mechanism

Cybercriminals evade detection by operating from jurisdictions with weak enforcement.

3.8 Impact of These Challenges

These challenges collectively result in:

- Low conviction rates
- High case backlogs
- Loss of public trust in digital platforms
- Increased financial and emotional harm to victims
- Threats to national security

A strategic, coordinated approach is required to overcome these barriers.

3.9 Conclusion

This chapter has examined the multiple challenges Bangladesh faces in combating cybercrime and managing digital evidence. The issues are not limited to legal deficiencies but are deeply rooted in technical capacity, institutional weaknesses, procedural inconsistencies, and socio-cultural barriers. Cybercrime's transnational nature further complicates enforcement efforts.

CHAPTER 4

LEGAL FRAMEWORK, ENFORCEMENT CHALLENGES, AND INSTITUTIONAL RESPONSES IN BANGLADESH

Introduction

As digital connectivity expands across Bangladesh, cyber-dependent and cyber-enabled offences have grown in frequency, complexity, and transnational reach. To respond to these threats, Bangladesh has introduced several laws, regulatory institutions, and enforcement units. However, questions remain regarding the adequacy, coherence, and constitutionality of the existing legal and institutional framework. This chapter examines the effectiveness of Bangladesh's legal responses to cybercrime, the treatment of digital evidence, and the enforcement challenges faced by investigative agencies, prosecutors, and the judiciary. It also evaluates how institutional arrangements—such as the Digital Security Agency (DSA), Bangladesh Telecommunication Regulatory Commission (BTRC), and cybercrime investigation units—operate in practice.

4.1 Existing Legal Framework Governing Cybercrime

4.1.1 The Information and Communication Technology Act, 2006 (as amended in 2013)

The ICT Act was Bangladesh's first dedicated legislation addressing cyber-offences. It criminalized hacking, data theft, identity misuse, fraud, publication of obscene material, and tampering with computer systems. The 2013 amendment increased penalties and made some offences non-bailable. Although pioneering, the Act faced criticism for vague definitions, harsh punishments, and potential misuse—leading to its gradual replacement by the Digital Security Act (DSA), 2018.

4.1.2 Cyber Security Ordinance, 2025 [5]

The DSA is currently Bangladesh's principal cybercrime legislation. It covers:

[5] [5] This ordinance, which was issued as a gazette in May 2025, replaces the previous Cyber Security Act of 2023 (CSA). The 2023 CSA had, in turn, replaced the controversial Digital Security Act (DSA) of 2018.

- Unauthorized access and hacking
- Spreading propaganda, false information, or defamation
- Cyber-terrorism
- Digital fraud and forgery
- Data breach and privacy violations

The Act grants authority to police to arrest without warrant under certain sections and allows the Digital Security Agency to oversee digital infrastructure protection. Critics argue that several provisions overlap with existing criminal laws, create broad discretionary powers, and may affect freedom of expression. Nonetheless, for cyber-security and digital evidence, the DSA remains the most used legal instrument.

4.1.3 Evidence Act, 1872 (amended 2025) and Admissibility of Digital Evidence

The 2022 amendment officially recognized:

- Electronic documents
- Emails, SMS, and call records
- Server logs, CCTV footage
- Computer-generated records

Challenges persist in ensuring chain of custody, authenticity, integrity checks (hashing), and expert testimony. Judges and lawyers often lack training in technical evaluation of digital evidence.

4.1.4 Bangladesh Telecommunication Act, 2001 and BTRC Regulations

BTRC monitors illegal mobile use, VoIP, SIM registration, and telecom fraud. It assists law enforcement by providing call detail records (CDR), IP logs, and telecommunication metadata. However, delays in data extraction and limited technical resources reduce investigative effectiveness.

4.1.5 Personal Data Protection Ordinance (PDPO) 2025[6]

Bangladesh is in the process of passing an Act on personal data protection legislation. The ordinance focuses on:

- Data collection and consent
- Data retention
- Cross-border data transfer
- Establishment of a Data Protection Office

If enacted, it will significantly influence the investigation of cyber offences and the admissibility of digital evidence.

4.2 Enforcement Mechanisms and Institutional Arrangements

4.2.1 Cyber Crime Unit of CID

The Criminal Investigation Department (CID) has a specialized Cyber Crime Unit responsible for:

- Email fraud investigation
- Social media harassment cases
- Financial cybercrime
- Digital forensics

CID operates multiple digital forensic labs but struggles with case overload, shortage of trained forensic examiners, and dependence on foreign expertise for advanced malware analysis.

4.2.2 Counter Terrorism and Transnational Crime (CTTC) Unit

CTTC focuses on cyber-terrorism, extremist online propaganda, darknet activity, and crypto-enabled financing. Their Cyber Intelligence Unit conducts surveillance and digital intelligence gathering. Lack of inter-agency coordination sometimes limits operational efficiency.

[6] The Personal Data Protection Ordinance PDPO 2025 provides a long-overdue structure for managing personal data.

4.2.3 Police Cyber Support for Women (PCSW)

PCSW provides online complaint mechanisms for cyber harassment, stalking, blackmail, and privacy violations, especially targeting women. It is one of the most accessible cybercrime reporting channels in Bangladesh. However, limitations include manpower shortages and slow response time for large-volume complaints.

4.2.4 Digital Security Agency (DSA) and e-Government Computer Incident Response Team (BGD e-Gov-CIRT)

These bodies focus on:

- Cyber-incident prevention
- National cyber-security risk assessments
- Sector-wise cyber-security audits
- Emergency response to cyber-attacks

CIRT provides threat intelligence, malware analysis, and CERT-to-CERT coordination, but lacks legal authority for criminal investigation.

4.3 Key Enforcement Challenges in Bangladesh

4.3.1 Technical Limitations and Resource Constraints

Many agencies lack:

- High-end digital forensic labs
- Trained cyber analysts
- Malware reverse-engineering tools
- Real-time monitoring systems
- Access to encrypted devices and cloud data

Most cyber investigations rely on basic forensic tools, limiting the ability to trace sophisticated cyber networks.

4.3.2 Procedural Challenges with Digital Evidence

Problems include:

- Improper chain-of-custody documentation
- Seizure without forensic imaging
- Difficulty verifying authenticity (hash mismatch)
- Limited judicial training
- Overdependence on foreign companies for data (Meta, Google, etc.)

Many cases collapse during trial because digital evidence fails legal standards.

4.3.3 Delayed Mutual Legal Assistance (MLA)

Cybercriminals often operate across borders. However, Bangladesh's requests for foreign data (cloud accounts, emails, social media records) face months-long delays due to:

- Lack of bilateral data-sharing agreements
- Non-compliance with international MLA standards
- Absence of a clear digital evidence request protocol

This significantly weakens prosecution.

4.3.4 Public Awareness and Underreporting

Victims often hesitate to report cybercrimes due to:

- Privacy concerns
- Social stigma
- Lack of confidence in law enforcement
- Fear of re-victimization

Underreporting makes cybercrime appear less severe than it truly is.

4.3.5 Overlapping Legal Provisions and Misuse Concerns

Critics argue that sections of the DSA overlap with:

- Penal Code (defamation)
- ICT Act (residual offences)
- Telecommunication Act

Broad discretionary powers may lead to inconsistent enforcement.

4.4 Institutional Prospects and Opportunities for Reform

4.4.1 Establishment of a National Cybercrime Coordination Center (NCCC)

A unified, inter-agency coordination body can:

- Streamline investigation
- Share digital forensic resources
- Maintain a national cyber-threat database
- Coordinate MLA requests

4.4.2 Introducing a Modern Digital Evidence Act

Bangladesh would benefit from legislation that clearly outlines:

- Standardized forensic processes
- Metadata retention obligations for telecom/ISPs
- Rules for cloud-based evidence
- Admissibility standards aligned with international best practices

4.4.3 Developing Capacity-Building Programs

Training judges, prosecutors, and law enforcement personnel is crucial. Government–university partnerships can develop digital forensic training programs.

4.4.4 Enhancing Public Awareness and Victim Support Mechanisms

Public awareness campaigns, school-level cyber safety programs, and victim counseling services can increase trust in institutions and improve reporting rates.

Conclusion

Bangladesh has made significant progress in developing a legal and institutional structure to combat cybercrime and manage digital evidence. The ICT Act, DSA, amended Evidence Act, and telecom regulations form a foundational framework. Specialized enforcement units—CID Cyber Unit, CTTC, PCSW, and CIRT—further strengthen national capability. However, systemic challenges such as inadequate forensic capacity, procedural gaps in evidence handling, limited cross-border cooperation, and overlapping laws reduce the effectiveness of cybercrime responses. To address these challenges, Bangladesh must adopt modern digital evidence standards, improve institutional coordination, and expand technical and human resources.

CHAPTER 5

CONCLUSION AND POLICY RECOMMENDATIONS

Introduction

This chapter provides a comprehensive synthesis of the research findings and outlines policy recommendations aimed at strengthening Bangladesh's capacity to combat cybercrime and effectively manage digital evidence. Building on the problems identified in the earlier chapters—legal gaps, institutional weaknesses, technical limitations, and socio-cultural barriers—this chapter proposes actionable solutions that can guide policymakers, law-enforcement bodies, and judicial authorities. As Bangladesh rapidly embraces digital transformation, robust cyber governance becomes essential to ensure national security, protect digital rights, and support sustainable digital economic growth.

5.1 Summary of Key Findings

5.1.1 Rising Cybercrime in a Digitally Evolving Landscape

Bangladesh is witnessing an exponential increase in cyber-dependent and cyber-enabled crimes. Online fraud, identity theft, social media harassment, financial scams, ransomware, and data breaches have become common. The country's growing digital ecosystem—mobile banking, e-commerce, and social media use—has created both opportunities and vulnerabilities. Lack of awareness and weak reporting culture intensify the problem.

5.1.2 Gaps in the Legal Framework

Although the Digital Security Act (DSA), ICT Act (2006), Evidence Act (amended 2022), and various telecom regulations provide a legal foundation, they suffer from:

- Overlapping provisions
- Broad discretionary powers
- Ambiguity in defining offences
- Insufficient procedural guidelines for digital evidence

- Limited adaptability to emerging cyber threats

Bangladesh lacks a comprehensive digital evidence law aligned with international standards such as the **Budapest Convention on Cybercrime**.

5.1.3 Weaknesses in Evidence Management and Forensics

Digital evidence handling is one of the weakest areas identified. Common issues include:

- Improper chain of custody
- Non-forensic seizure of devices
- Insufficient metadata preservation
- Inadequate forensic tools and training
- Lack of specialized digital evidence courts

Many cybercrime cases fail in court due to improper forensic documentation and lack of expert testimony.

5.1.4 Limited Institutional Capacity and Coordination

Despite having specialized agencies such as CID Cyber Unit, CTTC, PCSW, and CIRT, Bangladesh faces:

- Fragmented institutional structure
- Lack of inter-agency coordination
- Shortage of skilled personnel
- Dependence on foreign entities for cloud-based data
- Slow cross-border cooperation

5.1.5 Social and Cultural Barriers

Victims, especially women and children, often hesitate to report cyber offences due to fear, shame, or distrust of authorities. Public awareness regarding online risks, privacy, and data security remains low.

5.2 Policy Recommendations

5.2.1 Legal Reforms

1. Enact a Comprehensive Digital Evidence Act

A modern digital evidence law should include:

- Clear definitions and admissibility standards
- Digital chain-of-custody procedures
- Rules for cloud, blockchain, and encrypted evidence
- Mandatory hashing at seizure and analysis stages
- Standard forensic documentation formats

2. Repeal of the Digital Security Act by Cyber Security Ordinance 2025

Revisions should ensure:

- Clear definitions to reduce misuse
- Balanced enforcement powers
- Alignment with international cybercrime standards
- Distinction between genuine cyber offences and free-expression issues

3. Ratify the Budapest Convention

This would strengthen international cooperation, digital evidence sharing, and global cybercrime frameworks.

5.2.2 Institutional Strengthening

1. Establish a National Cybercrime Coordination Center (NCCC)

A unified body should coordinate:

- Forensic labs

- Intelligence sharing
- MLA requests
- Policy oversight

2. Create Dedicated Cybercrime Courts

Special cyber courts with trained judges and prosecutors would ensure faster and more consistent disposal of cases.

3. Expand Forensic Infrastructure

Investment is essential in:

- High-end forensic labs
- Mobile forensics hardware
- Malware analysis platforms
- Dark web monitoring tools

5.2.3 Capacity Building

1. Training for Law-Enforcement and Judiciary

Regular training should cover:

- Forensic acquisition and analysis
- Digital evidence documentation
- Encryption and cyber investigation techniques
- Cyber laws and international norms

2. Collaboration with Universities and International Partners

Academic institutions can host digital forensics programs, cyber labs, and research centers. International cooperation (INTERPOL[7], UNODC[8]) can help develop advanced expertise.

5.2.4 Public Awareness and Community Engagement

1. Nationwide Cyber Safety Campaigns

Schools, colleges, and workplaces should conduct training on:

- Online privacy
- Safe social media use
- Recognizing fraud and phishing
- Reporting cyber incidents

2. Strengthening Victim Support Services

Expand PCSW and introduce digital victim support hotlines, counseling services, and fast-response mechanisms.

5.2.5 Enhancing Cross-Border Cooperation

- Develop bilateral agreements with social media and tech companies for faster data access.
- Standardize Mutual Legal Assistance (MLA) processes.
- Participate in regional cyber task forces in South Asia.

[7] The International Criminal Police Organization
[8] United Nations Office on Drugs and Crimes

Conclusion

Cybercrime is a growing national security challenge for Bangladesh, driven by rapid technological expansion, rising digital dependency, and evolving criminal techniques. While Bangladesh has made significant progress through legislation such as the DSA and institutional arrangements such as CID Cyber Unit and CIRT, substantial gaps remain in legal clarity, forensic capability, institutional coordination, and public awareness.

A secure digital future requires **comprehensive legal reform**, **investment in forensic capacity**, **inter-agency integration**, and **widespread awareness programs**. Adopting international best practices and building specialized human resources will allow Bangladesh to address cyber threats effectively and harness the full potential of its digital transformation.

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